

Chapter 2

Characteristics of Video Material

We saw in the previous chapter that simply converting an analog video service into digital form results in an unacceptable increase in the bandwidth required to transmit the service even when sophisticated modulation schemes that transmit several bits/s/Hz are employed. For digital video to be a practical reality, it is essential that the number of bits required to represent each picture in a video sequence be significantly reduced. Fortunately, the characteristics of video material are such that substantial reductions in the number of bits can be achieved without noticeably affecting the subjective quality of the video service. In this chapter, we describe the characteristics that allow these savings to be made.

Figure 2.1 shows the first picture of the “Mobile and Calendar” video sequence that will be used to illustrate concepts as we consider the various signal processing techniques employed to compress video material. Longer versions of this sequence (among others) were used during the development of the MPEG digital standards that are at the heart of digital television. The picture contains 576 rows of pixels with each row containing 704 pixels. Although slightly less than CCIR Recommendation 601,¹ most of the development work for international standards was performed at this resolution.

2.1. PICTURE CORRELATION

Consider the picture shown in Figure 2.1, which is taken from the “Mobile and Calendar” sequence. Although this is an extremely “busy” picture, there are still large areas that are of a similar gray level. This includes the white background in the calendar, the light gray of the goat, the light background of the wallpaper, and the black of the body of the train. This “sameness” within a picture can be exploited to reduce the amount of data that needs to be transmitted to accurately represent the

¹ The missing pixels and rows of pixels are taken up by the horizontal and vertical blanking intervals.



Figure 2.1 First frame of the video sequence “Mobile and Calendar.”

picture. Let us take an extreme example. Consider a picture in which every pixel is the same shade of gray. In order to completely represent the picture all that would be needed would be the gray level of the first (top left) pixel together with the statement that every other pixel is the same shade of gray. The information about this one pixel is sufficient to allow the values of all the other pixels to be correctly determined.

Going to the other extreme, consider a picture made up of white noise. In this case, the value of every pixel needs to be individually specified because knowing the value of a particular pixel tells nothing about the value of any other pixel in the picture.

Mathematically, the “sameness” of a picture is measured by the autocorrelation function. This function measures how pixel “sameness” varies as a function of the distance between the pixels. The correlation coefficient r between two blocks of pixels $A(i,j)$ and $B(i,j)$ where i and j are the pixel positions within each block is defined as

$$r = \frac{\sum_i \sum_j (A(i,j) - \mu_A)(B(i,j) - \mu_B)}{\sqrt{\sum_i \sum_j (A(i,j) - \mu_A)^2 \sum_i \sum_j (B(i,j) - \mu_B)^2}} \quad (2.1)$$

where μ_A and μ_B are the mean values of $A(i,j)$ and $B(i,j)$, respectively.

For two blocks that are identical (e.g., any two blocks extracted from the picture where every pixel is identical), the correlation coefficient is one. For blocks that are completely uncorrelated (e.g., any two blocks extracted from the white noise picture),

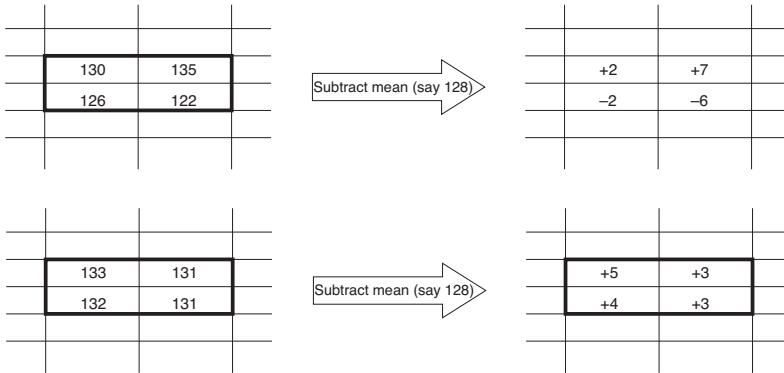


Figure 2.2 Calculation of correlation coefficient.

the correlation will be zero. In fact the correlation coefficient can take values in the range from -1 to 1 corresponding to the cases where $A(i,j) = -B(i,j)$ and $A(i,j) = B(i,j)$, respectively.

EXAMPLE 2.1

Consider the two 2×2 block of pixels shown in Figure 2.2. The mean of each block is first subtracted from each pixel in that block. Failing to do this leads to a correlation coefficient that is always positive and close to one irrespective of the pixel values because the mean value dominates the calculation.

$$r = \frac{(+2)(+5) + (+7)(+3) + (-2)(+4) + (-6)(+3)}{\sqrt{((+2)(+2) + (+7)(+7) + (-2)(-2) + (-6)(-6))((+5)(+5) + (+3)(+3) + (+4)(+4) + (+3)(+3))}}$$

$$= 0.0328$$

This corresponds to a pair of (admittedly small) blocks that are almost uncorrelated. Note that if the mean was not subtracted the correlation coefficient would be 0.9993. This clearly demonstrates how the mean can mask the underlying behavior within the block. ■

This idea can be easily extended to whole pictures using MATLAB.

EXAMPLE 2.2—MATLAB

Calculate the correlation coefficient for the luminance component of the first picture of the “Mobile and Calendar” sequence for horizontal shifts in the range from -10 to $+10$ pixels.

SOLUTION In order to obtain a reasonably global value for the correlation coefficient, it is necessary to use a large block of pixels. We use a 576×684 pixel block within the 576×704 pixel picture. This allows the block to be moved in the range from -10 to $+10$ pixels without running off the edge of the picture. This is shown in Figure 2.3.

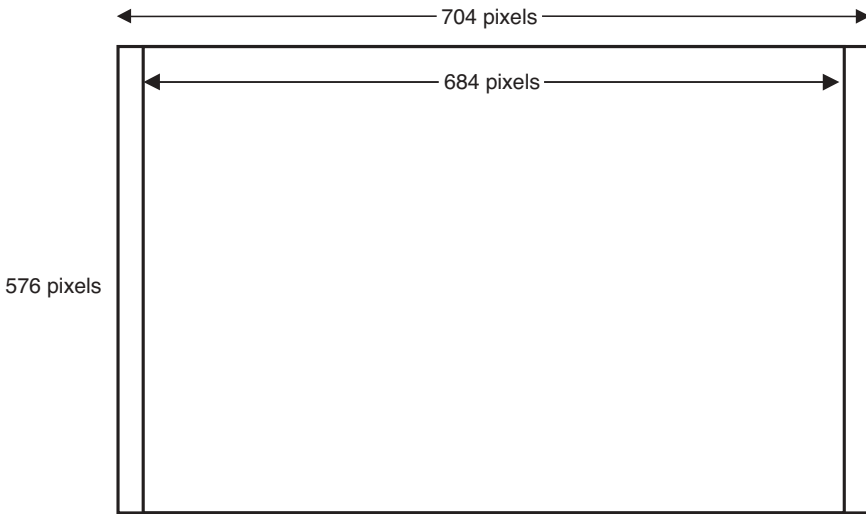


Figure 2.3 Block area for horizontal correlation coefficient calculation.

Assume that the array `calendar_1` contains the first picture of the “Mobile and Calendar” sequence. The appropriate MATLAB script is

```
horiz_corr = zeros(1,21);
[row,col] = size(calendar_1);
block = calendar_1(:,11:col-10);           %define block to be moved over picture
block = block - mean(mean(block));          %subtract block mean
for position = -10:10
    compare_block = calendar_1(:,11+position:col-10+position); %define block
    compare_block = compare_block - mean(mean(compare_block)); %subtract mean
    horiz_corr(1,position+11) = corr2(block,compare_block); % calculate coefficient
end
x=-10:10;
plot(x,horiz_corr);
grid
```

After appropriate labeling of the axes, the result is shown in Figure 2.4.

As expected, at a displacement of 0 the correlation coefficient is 1. Note that the correlation coefficient at a displacement of 1 pixel is, however, in excess of 0.9. Even at a pixel shift of 10 pixels, the correlation coefficient is still greater than 0.65. This clearly demonstrates that even for a complex picture such as this one, there is still a large amount of correlation that can be exploited to reduce the required data rate. ■

This idea can simply be extended to calculate a two-dimensional plot of the correlation coefficient. In this case the block of interest is moved over a range of -5 to $+5$ pixels in both the horizontal and the vertical directions with the result being plotted as a three-dimensional surface plot. After appropriate interpolation, the result is shown in Figure 2.5.

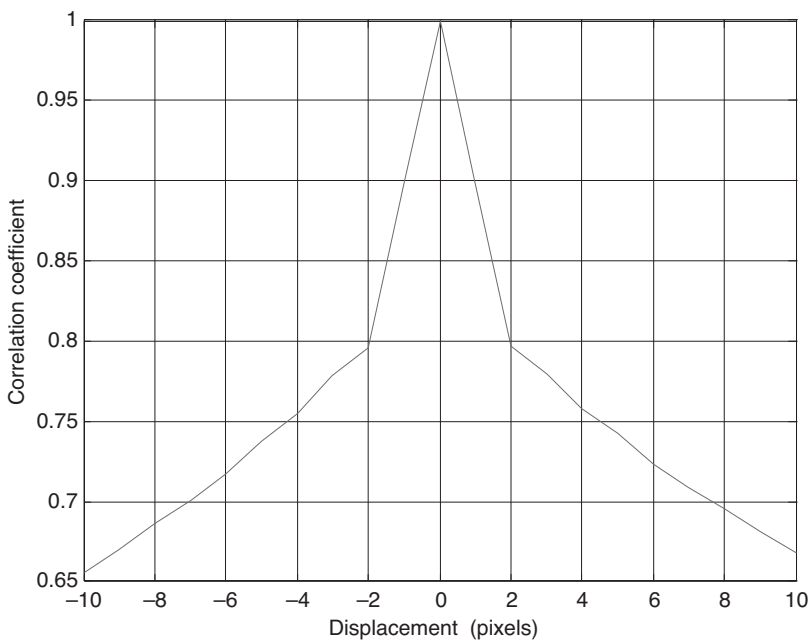


Figure 2.4 Horizontal correlation coefficient for luminance component of the first picture of sequence “Mobile and Calendar.”

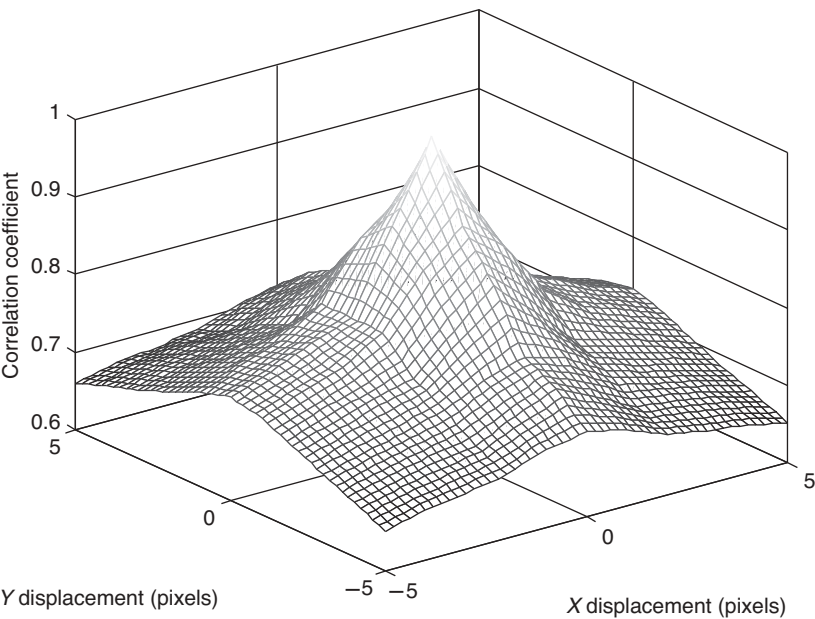


Figure 2.5 The three-dimensional surface plot of the correlation coefficient for the luminance component of the first picture of sequence “Mobile and Calendar.”

It is clear that correlation in all directions remains high for several pixel shifts from the pixel of interest. Even at a shift of 5 pixels left and 5 pixels down, the correlation is still as high as 0.65.

The conclusion from these examples is that if we know the value of a particular pixel then we also have a considerable amount of information about the values of nearby pixels. It turns out that correlation between adjacent pictures in a video sequence is as large, and often even larger, than is the correlation between pixels within a picture. How this information can be exploited to reduce the amount of data required to represent a picture is discussed in later chapters.

2.2. INFORMATION CONTENT

It is important to be able to quantify in some way the amount of information produced by a source. Consider the emission of a symbol “a” from a source with probability p_a . If $p_a = 1$ then symbol a is produced continuously by the source. There is therefore no surprise when it is produced and hence there is no information provided by its appearance.

On the contrary, if the various symbols produced by the source have a range of possibilities and, if the probability p_a is low, there is more surprise—and hence more information—in its occurrence. The smaller the value of p_a , the greater is the information content. The information content is related to the reciprocal of the probability of occurrence.

The amount of information I_a is defined as

$$I_a = \log_2 \left(\frac{1}{p_a} \right) = -\log_2 (p_a) \quad (2.2)$$

The base for the logarithmic function is in fact quite arbitrary. However, it is usual to use the base 2 as shown. In this case, the unit of information is the bit (a contraction of *binary digit*).

When p_a is equal to $1/2$, the information content is 1 bit. Thus 1 bit of information is sufficient to distinguish one of two equally likely events.

This definition of information content displays the following pleasing characteristics:

- $I_a = 0$ for $p_a = 1$. As stated earlier, if we are sure of the symbol prior to its occurrence then no information is gained.
- $I_a \geq 0$ for $0 \leq p_a \leq 1$. The occurrence of a symbol provides at worst no information but never brings about a loss of information.
- $I_a \geq I_b$ if $p_a < p_b$. Less likely events provide more information than more likely events.

- $I_{ab} = I_a \cap I_b = I_a + I_b$. If a and b are statistically independent events (i.e., if $p_{ab} = p_a p_b$) then the information content of the two events is the sum of the information content of each event.

If we consider a source producing m symbols $s_0, s_1, \dots, s_{m-1}$ with probabilities $p_0, p_1, \dots, p_{m-1}$, then the mean information content is given by

$$\begin{aligned}
 H &= E(I_{s_k}) \\
 H &= \sum_{k=0}^{m-1} p_k I_{s_k} \\
 H &= \sum_{k=0}^{m-1} p_k \log_2 \left(\frac{1}{p_k} \right) \\
 H &= - \sum_{k=0}^{m-1} p_k \log_2 p_k
 \end{aligned} \tag{2.3}$$

H is called the entropy of the source and measures the average information content per source symbol. The entropy provides a lower bound on the average number of bits required to transmit a sequence of independent symbols, given the probability of each symbol.

EXAMPLE 2.3

A source generates four symbols a, b, c, and d that have probability of occurrence of 0.4, 0.3, 0.2, and 0.1, respectively. What is the entropy of the source?

SOLUTION The entropy calculation is shown in Table 2.1.

Table 2.1 Entropy calculation for Example 2.3.

Symbol	p_i	$I = -\log_2(p_i)$	$-p_i \log_2(p_i)$
a	0.4	1.32	0.53
b	0.3	1.74	0.52
c	0.2	2.32	0.46
d	0.1	3.32	0.33
Total source entropy			1.84 bits/symbol

This process can be extended to measure the information content of a picture. In an 8-bit gray scale picture, each pixel takes one of 256 possible values (0–255). By calculating the probability of each of these 256 values, it is possible to calculate the entropy of the picture. Figure 2.6 shows the histogram of pixel intensities for the first luminance picture of the sequence “Mobile and Calendar.”

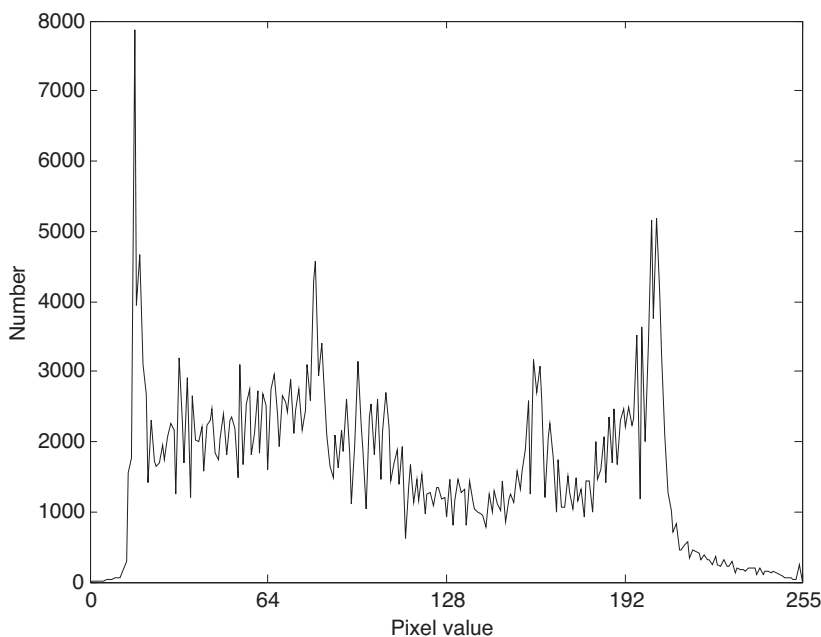


Figure 2.6 The histogram of the first luminance picture of the sequence “Mobile and Calendar.”

From Figure 2.6, it is clear that not all pixels are equally likely. When the picture entropy is calculated, it turns out that the entropy is 7.61 bits/pixel. This implies that the coding efficiency, defined as the entropy divided by the actual number of bits per symbol in the original picture ($7.61/8$ in this case), is already more than 95% efficient. Future chapters show that significantly more complex signal processing is needed to achieve the level of compression required for the successful introduction of digital television. If all 256 values were equally likely, the entropy would be 8 bits/pixel.

The key question now is how do we attempt to exploit the fact that the entropy is less than the number of bits per pixel currently required to represent the picture. This question can be answered by considering an old form of coding symbols (in this case letters of the alphabet)—Morse code. The Morse code representation of the 26 letters of the alphabet is given in Table 2.2.

The amount of time required to transmit a message is minimized by representing commonly occurring letters with a small number of keystrokes (e.g., the letter E) whereas less commonly occurring letters are represented by a larger number of keystrokes (e.g., the letters Q, Y, and Z). In Morse code, a dot (.) requires only one third of the transmission time of a dash (—). This implies that the number of keystrokes does not exactly correlate with transmission time. Nonetheless, the implications are clear. If we want to transmit symbols at the rate suggested by the entropy, we need to represent each symbol by a binary number with the same number of bits as the information content of the symbol. This is not always possible because the information content is not necessarily exactly an integer number of bits.

Table 2.2 Morse code for letters of the alphabet.

Letter	Morse code	Letter	Morse code	Letter	Morse code
A	.-	J	.-.-	S	...
B	-...	K	-.-	T	-
C	-.-.	L	.-..	U	..-
D	-..	M	-	V	...-
E	.	N	-.	W	.-.-
F	..-.	O	---	X	-.-
G	-.	P	.-.-	Y	-.--
H		Q	-.-.-	Z	--..
I	..	R	.-.		

EXAMPLE 2.4

A message consists of four symbols (a, b, c and d) with probabilities 0.75, 0.125, 0.0625, and 0.0625, respectively.

- Calculate the entropy of the message.
- Calculate the required number of bits per symbol and the coding efficiency if fixed length coding is employed.
- Calculate the required number of bits per symbol and the coding efficiency if variable length coding is employed.

SOLUTION (a) The entropy of the message can be calculated as shown in Table 2.3.

Table 2.3 Calculation of message entropy.

Symbol	p_i	$I = -\log_2(p_i)$	$-p_i \log_2(p_i)$
a	0.7500	0.415	0.311
b	0.1250	3.000	0.375
c	0.0625	4.000	0.250
d	0.0625	4.000	0.250
Total source entropy			1.186 bits/symbol

- As there are four symbols, representing each symbol by a fixed length code would require 2 bits/symbol as shown in Table 2.4.

Table 2.4 Representation with fixed length code words.

Symbol	p_i	Code word	Code word length
a	0.7500	00	2
b	0.1250	01	2
c	0.0625	10	2
d	0.0625	11	2

The average code word length (L) can then be calculated according to

$$\begin{aligned} L &= \sum_i (p_i)(\text{code word length}_i) \\ &= 0.75 \times 2 + 0.125 \times 2 + 0.0625 \times 2 + 0.0625 \times 2 \\ &= 2 \text{ bits/symbol} \end{aligned} \quad (2.4)$$

Given that every symbol is represented by 2 bits, this is hardly a remarkable result. The coding efficiency is defined below.

$$\text{Coding efficiency} = \frac{\text{Entropy}}{\text{Number of bits per symbol}} \quad (2.5)$$

In this case the coding efficiency is

$$\frac{1.186}{2.0} = 59.3\% \quad (2.6)$$

- (c) Now consider the variable word length coding shown in Table 2.5. The most likely symbol is represented by a short code word whereas less likely symbols are represented by longer code words.

Table 2.5 Representation with variable length code words.

Symbol	p_i	Code word	Code word length
a	0.7500	0	1
b	0.1250	10	2
c	0.0625	110	3
d	0.0625	111	3

In this case, the average code word length (L) can then be calculated according to

$$\begin{aligned} L &= \sum_i (p_i)(\text{code word length}_i) \\ &= 0.75 \times 1 + 0.125 \times 2 + 0.0625 \times 3 + 0.0625 \times 3 \\ &= 1.375 \text{ bits/symbol} \end{aligned} \quad (2.7)$$

The coding efficiency has been increased to $1.186/1.375 = 86.2\%$ a substantial improvement. ■

Details of techniques used to design variable length coding schemes will be considered in the next chapter.

2.3. THE HUMAN VISUAL SYSTEM

The previous two sections have discussed the mathematical properties of picture data that can be exploited to reduce the amount of data required to represent a picture. In considering the data compression problem, it needs to be firmly borne in mind that the aim of the process is to produce a reconstructed picture that,

when viewed by a human viewer, is of sufficient quality to meet the needs of the service. In particular, the aim is not to produce a reconstructed picture that is identical with the original picture that was imaged at the source. It follows that any parts of the picture that are not apparent to a viewer need not be preserved in the reconstructed picture. This allows huge savings to be made in the amount of data that needs to be transmitted. For these savings to be made, we need to have a reasonable understanding of the characteristics, and in particular the limitations, of the human visual system. These characteristics are discussed briefly in this section.

2.3.1. Perception of Changes in Brightness

According to Weber's law, the just detectable change in luminance (ΔY) is proportional to the luminance (Y). In fact

$$\frac{\Delta Y}{Y} \approx 0.02 \quad (2.8)$$

The response of the human visual system is therefore logarithmic. It also implies that a given change in luminance value due to a coding artifact is more visible in darker rather than light areas of the picture.

There is, however, another effect that needs to be considered. The luminance from a cathode ray tube (CRT) is related to the applied voltage (V) according to the nonlinear relationship.

$$Y = V^\gamma, \quad 2 \leq \gamma \leq 3 \quad (2.9)$$

When taken together, Weber's law and the nonlinear CRT relationship mean that the luminance perceived by a viewer is in fact approximately linearly related to the voltage applied to the CRT. Figure 2.7 shows a complete video compression system starting with a scene and camera followed by video coding (and decoding), and finally display on a CRT for the viewer. We have already seen that the combined effect of the CRT and the viewer's human visual system (as shown in the smaller rectangle in Fig. 2.7) is approximately linear. For the displayed image to look the

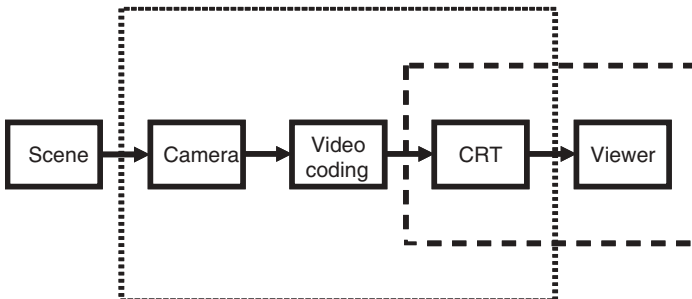


Figure 2.7 A complete video compression system.

same as the scene when viewed by the camera, the system contained in the larger rectangle from camera to CRT should also be linear. We want to perform video coding in the linear domain, and so video cameras invariably contain gamma correction circuitry.

$$V_{\text{out}} = V_{\text{camera}}^{\frac{1}{\gamma}} \quad (2.10)$$

2.3.2. Spatial Masking

It has been demonstrated experimentally that coding artifacts² in active regions of a picture (i.e., areas where there are sharp edges or other fast intensity changes) are less subjectively noticeable than those in inactive (i.e., flat) regions of a picture. This means that active areas of a picture can be coded more coarsely than inactive areas. Spatial masking is extensively exploited in modern video coders.

2.3.3. Temporal Masking

It has also been demonstrated that near scene changes in a video sequence there is a significant reduction in the subjective impact of coding artifacts. This means that pictures immediately after a scene cut (and to a lesser extent prior to the scene cut as well) can be coded more harshly than other pictures in the sequence without having any significant impact on subjective service quality.

2.3.4. Frequency Sensitivity

Figure 2.8 shows the approximate spatial frequency sensitivity of the human visual system for luminance information. It is clear that the frequency sensitivity is highest at low and medium frequencies and decreases rapidly at high frequencies. This does not mean that all high-frequency information can be deleted because edges introduce significant amounts of such high-frequency data that, if removed, would lead to a general blurring of the reconstructed image. Rather it means that high-frequency information can be represented less accurately than lower frequency information. This aspect of the human visual system is also widely exploited in modern video coders.

It is also worth noting that the contrast sensitivity for chrominance (color) information falls off at significantly lower frequencies than is the case for luminance

² The representation of parts of a picture can be made more and more approximate, or “coarse,” if fewer bits are available to represent it. In such situations, artifacts of the coding occur as either visible or invisible distortion.

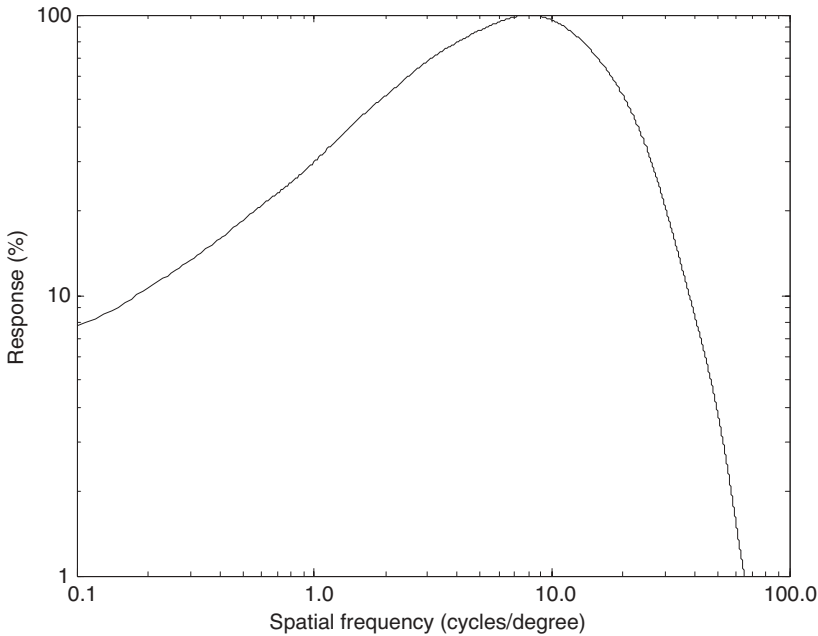


Figure 2.8 Luminance contrast sensitivity response as a function of frequency for the human visual system.

information. For this reason, chrominance information is invariably sampled in the spatial domain at a lower sampling rate than the luminance information.

2.3.5. Tracking of Motion

When an object in a video sequence first begins to move, there is a short period of time before the eye starts to track the motion. The eye's ability to resolve fine spatial detail in the object is reduced during this period. This is a similar effect to that described earlier at scene cuts. As an object continues to move, the eye's ability to track this detail improves providing that the motion is sufficiently slow that the eye can track it. Coding artifacts near objects that have just begun to move are therefore less noticeable than in other areas of the picture.

2.3.6. Conclusion

A thorough understanding of the human visual system is essential in the design of effective video encoders. Much of the compression achieved relies upon the introduction of coding artifacts in areas of a picture where they will not be subjectively noticeable. Although some conclusions have been drawn about the

characteristics of the human visual system in this section, this is still an area of active research.

2.4. SUMMARY

In this chapter, we have introduced the characteristics of images and the human visual system that can be exploited to allow the efficient representation of video material. In the next two chapters, we introduce the signal processing techniques that have been developed to facilitate this exploitation.

PROBLEMS

- 2.1 A progressive television service called 480p is to be introduced with each picture being made up of 720 pixels/line and 480 lines/picture. The picture rate is 60 pictures/s. Each pixel requires 12 bits of data to represent luminance and chrominance information. Calculate the efficiency required from the modulation scheme (in bits/s/Hz) so that the digitized video service can be carried in a channel with bandwidth 6MHz.
- 2.2 Repeat Problem 2.1 for a 1080i service that is made up of 2:1 interlaced pictures transmitted at a picture rate of 30 pictures/s. Each picture (i.e., a pair of fields) contains 1440 pixels/line and 1080 lines/picture.
- 2.3 Calculate the correlation coefficient between the 2×2 pixel blocks shown in Figure 2.9. Assume that the mean value of all pixels in the picture is 128.

146	123		157	117
130	105		140	91

Figure 2.9 Pixel data for Problem 2.3.

- 2.4 Calculate the correlation coefficient between the 3×3 pixel blocks shown in Figure 2.10. Assume that the mean value of all pixels in the picture is 128.

205	130	247		217	172	174
163	224	188		134	215	97
153	185	182		52	5	213

Figure 2.10 Pixel data for Problem 2.4.

- 2.5 What areas of a picture would you expect to display high correlation values and which areas would you expect to display low correlation? Consider the first picture of the “Mobile and Calendar” sequence. What areas of the picture will display local high and low values of correlation?
- 2.6 Consider the first picture of other video sequences. What areas of each picture would you expect to display local high and low values of correlation?
- (a) Explain how this could be calculated.

- (b) Under what circumstances would you expect high and low values of temporal correlation?

2.7 As indicated in the chapter, the correlation coefficient can take values in the range from -1 to $+1$. Consider the 4×4 pixel block shown in Figure 2.11 that has been drawn for a set of pixel values with mean value 128.

150	145	161	168
144	153	170	101
141	155	148	84
145	160	160	92

Figure 2.11 Pixel data for Problem 2.7.

- (a) Calculate the 4×4 pixel block that has a correlation coefficient of $+1$ when compared to this block.
- (b) Calculate a 4×4 pixel block that has a correlation coefficient of 0 when compared to this block.
- (c) Calculate a 4×4 pixel block that has a correlation coefficient of -1 when compared to this block.
- 2.8** A source generates four symbols a, b, c, and d with probabilities 0.50, 0.20, 0.20, and 0.10, respectively. What is the entropy of the source? If each symbol is represented by a fixed length binary code word, what is the coding efficiency?
- 2.9** A source generates six symbols a, b, c, d, e, and f with probabilities 0.30, 0.20, 0.20, 0.15, 0.10, and 0.05 respectively. What is the entropy of the source? If each symbol is represented by a fixed length binary code word, what is the coding efficiency?
- 2.10** A source produces three symbols with probabilities 0.90, 0.05, and 0.05, respectively. What is the entropy of the source? If each symbol is represented by a fixed length binary code word, what is the coding efficiency?
- 2.11** The source described in Problem 2.10 has symbols grouped in pairs prior to encoding. Each symbol can be assumed to be independent of any other symbol. What is the entropy of the source now? If each pair of symbols is represented by a fixed length binary code word, what is the coding efficiency?
- 2.12** Repeat Problem 2.11 for the case where the symbols are grouped in threes.
- 2.13** A source generates three symbols. One symbol has probability p whereas both of the other two symbols have the same probability. Plot source entropy as a function of the probability p .
- 2.14** The coding efficiency using variable length code words is usually less than 100%. Under what circumstances will this coding efficiency be exactly 100%?
- 2.15** When variable word length code words are used to represent symbols of differing probabilities, it is usual to ensure that no short code word is a prefix for a longer code word (i.e., if one code word is 00 then 001 could not be another code word as it begins with 00). Explain briefly why this might be an advantage. For Problems 2.8 and 2.9 suggest appropriate variable length code words that fulfill this requirement.

- 2.16 The Morse code is not a code where short code words cannot be a prefix of longer code words (e.g., the letter E is represented by a single dot whereas the letter I is represented by two dots). When transmitting Morse code, the operator needs to pause between letters to allow correct decoding at the receiver. What would be the impact of using such a code in a digital television system?
- 2.17 The length of variable word length codes is designed to match the long-term probabilities of each of the symbols generated by the source. Although this should lead to a saving overall, it is possible in some instances that the number of bits generated to represent a string of symbols can be more if the symbols were encoded using fixed length code word. Explain the circumstances where this is possible.
- 2.18 Four symbols are represented by the variable length code words as shown in Table 2.6. A string of transmitted symbols consists of the symbols A,B,A,B,C,D,A,A, respectively.

Table 2.6 Variable length code words for Problem 2.18.

Symbol	Code word
A	0
B	10
C	110
D	111

- (a) Calculate the generated bit stream corresponding to this run of symbols.
- (b) The second bit of this bit stream is received in error. Decode the received bit stream and comment on the result.
- (c) Suggest a method that might be used to overcome the problem of transmission errors when variable length coding is used.
- 2.19 The characteristics of the human visual system are very complex. Over the years, researchers have developed tests that attempt to quantify these characteristics. Design an experiment that might allow you to verify Weber's law.
- 2.20 Repeat Problem 2.19 but this time design an experiment that measures the frequency response of the human visual system.

MATLAB EXERCISE 2.1: CORRELATION COEFFICIENT WITHIN A PICTURE

The text has discussed in some detail the idea of correlation within a picture. In this exercise, you use MATLAB to measure the correlation coefficient of pictures from different video sequences. You also study the variation in the local value of the correlation coefficient within an image.

Section 1 Correlation coefficient of pictures from different sequences

- (a) For the first picture of a video sequence, calculate the two-dimensional correlation coefficient over the range of ± 5 pixels. Plot the result as a

three-dimensional plot. Your final result should look very similar to the plot shown in Figure 2.5. Note that the readability of the plot can be increased by interpolating between integer points using the two-dimensional interpolation function available in MATLAB.

- (b) Repeat this exercise for the first picture of other video sequences. Comment on the results. Which pictures would you expect to be easy to compress? Which pictures would you expect to be more difficult to compress? Why?

Section 2 Correlation coefficient within a picture

- (a) Extend the MATLAB program developed in the previous section so that you can select individual 16×16 pixel blocks within a picture. Choose different blocks within the first picture of your sequence. Calculate the two-dimensional correlation coefficient for each of these blocks again in the range of ± 5 pixels. You might find it helpful to highlight the selected block in some way (such as surrounding it with a white border). Use your program to determine what picture characteristics give rise to high and low values for the correlation coefficient.
- (b) Use your program to perform a similar study on the first picture from other video sequences. Comment on whether your observations for these pictures correspond to those for the first sequence.

MATLAB EXERCISE 2.2: CORRELATION COEFFICIENT BETWEEN PICTURES IN A SEQUENCE

As well as being able to calculate the correlation coefficient within a picture, it is also possible to calculate the correlation coefficient between pictures in a sequence. This is studied in this exercise.

- (a) Load the third picture from a video sequence into MATLAB. Now calculate the temporal correlation coefficient with the first five pictures in the sequence. Plot the correlation coefficient as a function of temporal displacement over the range of ± 2 frame times.
- (b) Repeat this calculation for other video sequences. Plot these on the same graph as the one used for the first sequence.
- (c) What characteristics of a sequence lead to high and low values of temporal correlation?

MATLAB EXERCISE 2.3: ENTROPY OF A PICTURE

In this exercise we plot the histogram of a picture. We then calculate the entropy of the picture and the coding efficiency when represented by fixed length code words.

Section 1 Histogram of a picture

- (a) For the first picture of a video sequence, calculate the number of pixels at each of the possible 256 gray scale values (range 0–255). Plot the histogram.
- (b) Calculate the histogram for the first picture of other sequences.
- (c) Comment on any differences between the histograms. Which picture would you expect to have the highest entropy? Which picture would you expect to have the lowest entropy?

Section 2 Calculation of entropy

- (a) Extend the program developed in the previous section to calculate the entropy (H) of each picture using the formula

$$H = - \sum_{k=0}^{m-1} p_k \log_2 p_k \quad (2.11)$$

where p_k is the probability of a pixel having intensity k .

Note that there may be some gray scale values that have a probability of zero. Your program needs to be able to deal with this situation.

- (b) How do the actual entropy values compare with the estimate you made in the previous section?